

Jack A. Logan



Research Profile

Theoretical and computational physicist with a PhD and 9+ years of experience working at the intersection of statistical mechanics, geometry, and machine learning. My research focuses on emergent structure, representation learning, and physically grounded modeling, with applications spanning soft matter, polymers, proteins, and interpretable neural networks. I work across theory, simulation, and computation to develop models that reveal underlying mechanisms and structure.

Selected Research Contributions

Independent Research

Participation Ensembles: A Statistical Framework for Interpretable Neural Representations (PyTorch) (In Progress)

- Developed a **general statistical framework** for analyzing neural representations from activation data using a single event-mechanism participation matrix, unifying superposition, polysemanticity, feature usage, and interactions within a single probabilistic object.
- Introduced **entropy- and interaction-based diagnostics** to quantify feature mixing, reuse, and circuit structure, enabling principled identification of redundant versus complementary mechanisms in trained neural networks.
- Demonstrated how these quantities can be **used as training regularizers and response metrics**, providing interpretable control over representation structure and a physics-inspired perspective on representation learning.

Multi-Agent Communication: Guide and Explorer (Gymnasium, PettingZoo, Stable-Baselines3) (In Progress)

- Developing a toy environment where an explorer must traverse a hazardous grid while a guide communicates safe paths through a noisy channel.
- Designed to test **coordination, emergent communication, and collective problem solving**.

Emergent dihedral symmetry in a 1-layer Transformer (PyTorch)

- Trained and reverse-engineered a **minimal 1-layer Transformer to learn dihedral group D_3 actions** without explicit supervision on compositions; designed causal circuit-composition interventions to test whether learned operator representations closed under group multiplication.
- Applied sparse autoencoders and geometric analysis (activation patching, cosine similarity, PCA) to **uncover feature superposition and aliasing in learned representations**, identifying a failure mode where non-abelian structure collapses into partially commutative latent embeddings.
- Connected empirical interpretability results to representation theory, showing how vector-based operator encodings break compositionality even for simple groups. Delivered a **concrete mechanistic explanation** for why single-step accuracy can coexist with systematic multi-step reasoning failures.

Surrogate Neural Network Force Field (PyTorch)

- Developed a PyTorch neural network surrogate force field for Lennard-Jones particles, using radial basis function features and a central-force assumption to predict accurate particle interactions.
- Implemented full training pipeline with feature engineering and force reconstruction, demonstrating how physical priors and physics-informed representations can simplify ML models.

- Visualized results in Blender and benchmarked against standard MD simulations.

Jamming On Curved Manifolds (C++, RATTLE)

- Designed and implemented a C++ MD engine for simulating polydisperse particles constrained to curved surfaces with variable radii (spheres, tori, cylinders, and planes).
- Built custom meshing algorithms to measure packing fractions and study jamming transitions on 2D curved manifolds.

Academic Research (PhD/Postdoc)

Structural properties of folded proteins (molecular dynamics, C++, Python)

- Developed an object-oriented C++ molecular dynamics framework to simulate multiple levels of coarse-grained protein folding.
- Ran large simulation ensembles on HPC clusters to study emergent structural features across folding trajectories.
- Analyzed simulation outputs to connect interaction rules with observed folding behavior.
- Publication chosen as Editors' Suggestion.

SymBOPs: Symmetrized Bond Order Parameters (group theory, tensors, symmetry representations, computational validation)

- Developed symmetry-specific bond order parameters (BOPs) grounded in group theory to characterize local structure beyond traditional BOPs.
- Partnered with experimental and computational teams to validate the framework on self-assembled systems, revealing symmetry-defined domains, including structurally disordered but symmetry-coherent regions.
- Resulted in three peer-reviewed publications, including two Editor's Picks.

Geometric and topological entropies of sphere packing (statistical mechanics, Monte Carlo)

- Developed a statistical-mechanical framework to quantify entropy in random sphere packings that generalizes concepts of granular and glassy configurational entropies for the case of non-jammed systems.
- Designed custom Monte Carlo algorithm to generate and sample constrained packing ensembles.
- Used computational experiments to build constrained sphere packings and relate packing geometry and geometric and topological entropy.

Compact interaction potential for van der Waals nanorods (analytical framework, asymptotics)

- Derived compact analytical expressions for van der Waals interactions between nanorods in arbitrary relative configurations.
- Enabled efficient modeling of anisotropic self-assembly without full-scale particle simulations.
- Provided immediately usable interaction models for computational studies of nanorod assemblies.

Professional Experience

Yale University, New Haven, CT

Postdoctoral Fellow, Mechanical Engineering & Materials Science

Advisor: Corey O'Hern

2022-2025

Brookhaven National Laboratory, Upton, NY

Research Assistant, Center for Functional Nanomaterials

Advisor: Alexei V. Tkachenko

2016-2022

Selected Publications

- Logan, J.A., et al. (2025) *Effect of stereochemical constraints on the structural properties of folded proteins*. Physical Review E, *Editors' Suggestion*.
- Logan, J.A., et al. (2023). *Symmetry-specific characterization of bond orientation order in DNA-assembled nanoparticle lattices*. Journal of Chemical Physics, *Editors' Pick*.
- Logan, J.A., Tkachenko, A.V., (2022). *Geometric and Topological Entropies of Sphere Packing*. Physical Review E.
- Mushnoori, S., Logan, J.A., et al. (2022). *Controlling morphology in hybrid isotropic/patchy particle assemblies*. Journal of Chemical Physics, *Editors' Pick*.
- Logan, J.A., et al. (2022). *Symmetry-specific orientational order parameters for complex structures*. Journal of Chemical Physics.
- Logan, J.A., Tkachenko, A.V., (2018). *Compact interaction potential for van der Waals nanorods*. Physical Review E.

Education

Stony Brook University, Stony Brook, NY

PhD in Physics, *2022*

Advisor: Alexei Tkachenko

Dissertation: *Geometric and Topological Aspects of Self-Assembly: from spheres and rods to designer particles*

Stony Brook University, Stony Brook, NY

Master of Arts in Physics

Long Island University, Brookville, NY

Master of Science in Applied Mathematics

Summa Cum Laude

Research Tools

Programming & HPC: Python, C++, CUDA, parallel computing on HPC clusters

Modeling & Simulation: Molecular Dynamics, Monte Carlo methods

Machine Learning: PyTorch, neural networks, physics-informed ML, mechanistic interpretability

Data & Collaboration: statistical analysis, optimization, Git/GitHub

Leadership & Awards

- Mentor, student Summer projects, Yale University (2024)
- GSNP Postdoctoral Speaker Award Finalist, APS March Meeting (2024)
- Team Leader, Protein Folding Simulation Project, Yale University (2022-2025)
- PEB Travel Fund Award, Yale University (2023)
- Postdoctoral Scholars Travel Fund Award (2023)
- Coordinated interdisciplinary collaborations to computationally and experimentally validate the SymBOPs theoretical framework (2021)
- Jonathan Kaufman Student Excellence Prize (2019)